

# **TfL What-If Simulator**

## **User Guide and Demonstration Script**

Step-by-step guide for using and presenting the implemented system

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# 1. Audience and Purpose

This document explains how to use the implemented TfL What-If Simulator and provides a structured demonstration script. It focuses on the visible system behaviour and how to present it clearly.

# 2. Quick Start

1. Open the web application and wait for the Underground map to load.
2. Use the collapsed search control if you want to jump to a specific station.
3. Click one station to set a route start, then click another station to set the destination.
4. Use the main-screen What-If toggle to enter simulation mode.
5. Close stations or lines, then plan a route again to show how the network adapts.
6. Save the scenario if you want to reload the same disruption state later.

# 3. Using the Map

The map is the main workspace. Users can pan and zoom like a normal web map. Stations and Underground connections are rendered as interactive map elements. Station markers can be clicked for route planning or What-If closure interaction depending on the current mode. A HUD/control area provides reset and route-management actions.

| Action                        | Expected behaviour                                                                   |
|-------------------------------|--------------------------------------------------------------------------------------|
| Pan map                       | Move around the London network while keeping the map constrained to the useful area. |
| Zoom map                      | Increase or decrease map detail. Station marker sizing adapts to zoom level.         |
| Click station in route mode   | Select start or destination station depending on current route state.                |
| Click station in What-If mode | Toggle closure state or use contextual route actions depending on interaction mode.  |
| Reset view                    | Return the map and route state to a clean baseline.                                  |

# 4. Planning Routes

To plan a route, select a start station and then select a destination station. The system calculates a path through the graph and highlights it on the map. The route panel summarises the route, estimated travel time and interchange information. After the route is found, the map adjusts so the selected route can be seen clearly.

The route algorithm uses travel time weighting and a penalty for line changes. This means it is not simply counting the number of stations; it is trying to produce a more realistic route by considering travel time and the cost of changing line.

# 5. Using What-If Mode

What-If mode is the simulation mode. It allows the user to model hypothetical disruptions without changing the live network graph. When enabled, simulated closures take priority over live closure state. The mode is intentionally visible from the main screen so users do not have to search through the sidebar to find it.

| What-If action        | Purpose                                         | Result                                                                     |
|-----------------------|-------------------------------------------------|----------------------------------------------------------------------------|
| Enable What-If toggle | Enter simulation mode.                          | The UI changes to a simulation context and closure controls become active. |
| Close station         | Model a station outage or inaccessible station. | The station is marked as closed and route planning avoids it.              |
| Close line            | Model a full line suspension.                   | Routes ignore all edges belonging to that line.                            |
| Reset closures        | Return simulation to no user-defined closures.  | Closed station and line sets are cleared.                                  |
| Disable What-If       | Return to live/normal mode.                     | Simulation closures no longer affect route planning.                       |

## 6. Saving and Loading Scenarios

A scenario is a named snapshot of the current What-If closures. It stores closed station IDs, closed line IDs and metadata such as creation/update time. Scenarios are saved in the browser using localStorage as JSON. This means they persist after refresh on the same device/browser, but they are not synced to other devices.

7. Turn on What-If mode.
8. Close one or more stations or lines.
9. Enter a scenario name.
10. Save the scenario.
11. Refresh or continue using the app.
12. Select the saved scenario to load it again.

This approach is suitable for assessment because it demonstrates persistence without requiring user accounts, authentication or additional database schema. It also protects the Neo4j transport graph from being modified by temporary simulations.

## 7. Using Live Data Features

When What-If mode is off, the application can display live service context from TfL APIs. This includes line status/delay information, live or fallback train movement and live station closure awareness. If TfL data is unavailable, the application should continue to operate with fallback values rather than failing completely.

| Feature           | How to demonstrate it                                                                    |
|-------------------|------------------------------------------------------------------------------------------|
| Line status       | Open the controls and show the status labels beside Underground lines.                   |
| Delay information | Hover/tap a line with status details to show reason/additional information if available. |
| Train display     | Enable train visibility and show markers moving along the network.                       |
| Feed status       | Point out whether the display is using live TfL arrivals or schedule fallback.           |

## 8. Suggested Demonstration Script

| Demo stage        | What to do                                                                                                               |
|-------------------|--------------------------------------------------------------------------------------------------------------------------|
| Opening statement | This is an interactive TfL What-If Simulator that combines a graph-based Underground model with live service context and |

|                         |                                                                                                                                                   |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
|                         | hypothetical disruption modelling.                                                                                                                |
| Normal route planning   | I will select a start station and destination station. The system calculates a route using weighted graph traversal and highlights it on the map. |
| Route visibility        | After selecting the route, the camera adjusts to show the route clearly, reducing manual panning and zooming.                                     |
| What-If discoverability | Based on questionnaire feedback, the What-If toggle has been moved onto the main map screen so users can find it quickly.                         |
| Station closure         | I will close a station and rerun the route. The route now avoids the closed station where possible.                                               |
| Line closure            | I will close a full line. This removes that line from the routing options and demonstrates network disruption simulation.                         |
| Scenario persistence    | I will save this disruption as a scenario, reset/change the state, and then reload the scenario from browser storage.                             |
| Live context            | Outside What-If mode, the application can show live line status and train movement with fallback handling if TfL data is unavailable.             |
| Closing statement       | The system demonstrates map interaction, graph routing, live data integration, user-centred UI iteration and lightweight scenario persistence.    |

## 9. Troubleshooting

| Issue                      | Likely explanation                                                   | What to check                                               |
|----------------------------|----------------------------------------------------------------------|-------------------------------------------------------------|
| Map does not show stations | /api/stations failed or Neo4j connection issue.                      | Check Neo4j environment variables and API response.         |
| Routes do not calculate    | Start/end not selected, graph not loaded or disruptions block route. | Clear route, reset closures and try two connected stations. |
| Saved scenarios disappear  | Browser storage cleared or different browser/device used.            | Check same browser and localStorage availability.           |
| Train feed shows fallback  | TfL arrivals unavailable or insufficient trackable vehicle IDs.      | This is expected fallback behaviour.                        |
| Controls overlap           | Viewport too small or layout offsets need adjustment.                | Test desktop and mobile breakpoints.                        |